Name ____ Date ___________ Period _____

Domain and Range from Problem Situations

Algebra II | Unit 1 | Day 2 | TEKS A2.7.I, A2.2.A

Objective: I can find the domain and range of a real-world situation, explain what they mean in context, and tell the difference between the domain of a function and the domain that the situation allows.

Part 1 — Name the Variables

Which quantity is the input and which is the output?

1. Hours you work and the money you earn
   independent: ______________________ dependent: ______________________
2. Number of tickets bought and the total cost
   independent: ______________________ dependent: ______________________
3. Seconds since a ball was thrown and its height
   independent: ______________________ dependent: ______________________

Part 2 — Function Domain vs. Contextual Domain

A cube-shaped box has side length x inches. Its volume is **$V(x) = x^3$**.

What domain does the **equation** x^3 accept? ________________________________

What domain does the **box** allow? ________________________________

Why? ____

Big idea: the equation may accept numbers the ______________ does not. Always ask what makes sense.

Part 3 — Domain and Range in Context

Give the contextual domain and range. Use an inequality and interval notation.

Example 1. Concert tickets cost \$12 each. You may buy at most 8. Total cost $C(t) = 12t$.

Domain: ____ Discrete or continuous? ______________

Range: ____

You Try 1. A ball is thrown. Its height is $h(t) = -16t^2 + 64t$ feet, and it is in the air from $t = 0$ to $t = 4$ seconds. Its greatest height is 64 feet.

Domain: __________________ interval ______________ Discrete or continuous? ______________

Range: __________________ interval ______________

You Try 2. A parking garage charges \$3 per hour and is open 6 a.m. to 10 p.m. (16 hours).

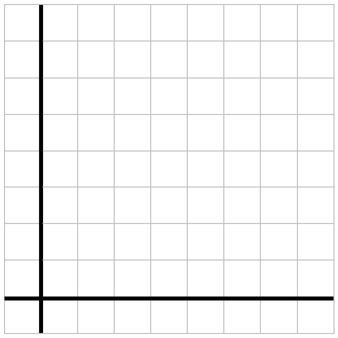
Domain: ___________________________ Range: ___________________________

Part 4 — Reading a Situation from a Graph

Each square is 1 unit. Plot the points, then answer in context.

A water tank is drained. Gallons g after m minutes:

m	0	1	2	3	4
g	8	6	4	2	0



Answer in context.

Domain: ____________________

Range: ____________________

What does the domain mean here?

____

What does a range of 0 mean?

____

Part 5 — Does the Situation Restrict the Domain?

Write yes or no, and if yes, give the restriction.

1. The side length of a square garden ___________________________
2. The temperature in a freezer over one day ___________________________
3. The number of buses needed for a field trip ___________________________

Practice

Give the contextual domain and range for each. Show your thinking on the lines.

1. A phone plan costs $30 per month plus $5 per gigabyte, up to 20 GB.

____

Domain ___________________________ Range ___________________________

2. A theater has 250 seats. Tickets are $9 each, and t tickets are sold.

____

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Domain ___________________________ Range ___________________________

3. A diver descends from the surface to 40 meters below sea level.

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Range of depth ___________________________ Why negative? ____________________

Exit Ticket

A taxi charges a $4 flat fee plus $2 per mile, for trips up to 10 miles. Cost $C(m) = 4 + 2m$.

Domain ___________________________ Range ___________________________

Why does the range start at 4, not 0? ___________________________________